Minimum Extendable Fano Projections and the Five-Cycle Double Cover Conjecture

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Resolution status. The five-cycle double cover conjecture remains unresolved. This note proves an exchange theorem, settles the minimum-support cases through size fifteen for finite connected bridgeless loopless cubic multigraphs (parallel edges allowed), and gives exact finite censuses. It proves neither the conjecture nor its negation. The standard conjecture for finite bridgeless multigraphs reduces to the cubic setting by Proposition 1.1.

Abstract

Hušek and Šámal recently expressed the five-cycle double cover conjecture as the problem of selecting a nowhere-zero $\mathbb{F}_2^3$-flow with a component-parity property. We study the support h of one binary coordinate of such a flow. Call h *extendable* if it occurs as a coordinate of some nowhere-zero $\mathbb{F}_2^3$-flow.

We prove a simultaneous exchange theorem for a minimum-cardinality extendable projection. Write the flow as $f = (h, s)$, with $s : E \to \mathbb{F}_2^2$, and partition h into the four affine value classes $M_c = \{e \in h : s(e) = c\}$. For every c, the projection h is a minimum-cardinality binary cycle containing M_c. Equivalently, every binary cycle C avoiding M_c satisfies

$$|C \cap h| \le |C - h|.$$

Thus every circuit component of h contains all four affine colours, and every support-shortening cycle meets all four classes. The proof is the addition of the Fano value $(1, c)$ along C.

We first prove, without computation, that every globally minimum extendable projection of size at most five is cleanable. We then give a stronger direct boundary repair. Independently on every component of $G - h$, apply an element of $\mathrm{GL}(2, 2)$ to the low-flow values; on every circuit of h, solve the resulting cyclic vertex equations while allowing zero low values, since the first coordinate is one there. Cleanliness becomes four explicit cut-parity equations per component. An exact affine/dihedral quotient exhausts every support shape, proper four-colour circuit word, and component partition through total support size ten. All 125,178 valid dirty canonical states admit a direct repair. At size eleven the same test first fails: there are 14 failed $5 + 6$ boundary states, in 11 full symmetry orbits. For every failure, however, an explicit boundary certificate makes the six-circuit low-valued and nowhere zero, so deleting it from the first-coordinate support produces an extendable projection of size five. Global minimality excludes all 14. Thus every minimum extendable projection of size at most eleven is cleanable. At size twelve, 13,788,824 dirty boundary states either clean directly or admit a strict circuit deletion; two independent enumerators agree exactly. Thus every minimum extendable projection of size at most twelve is cleanable. At size thirteen two independently written exact classifiers agree on all five support shapes and all 189,998,862 charge-valid states. Of the 159,369,966 dirty states, 159,362,292 clean directly and 7,674 admit a strict circuit deletion; there are zero residuals. Thus every minimum extendable projection of size at most thirteen is cleanable.

At size fourteen the primary exact classifier enumerates 9,481 word orbits and 2,616,134,989 charge-valid states. Of the 2,255,478,176 dirty states, 2,255,331,588 clean directly and 146,364 admit a strict circuit deletion. The 224 residuals all have shape $7+7$. A separately written full $7+7$ enumerator independently reproduces all 333 word orbits, 91,481,505 charge-valid states, and the identical residual set. We prove a two-colour path-switch lemma for the split-occurrence boundary model and give 724 literal deletion rows that meet every possible four- or six-terminal path pairing. Every residual therefore has a strict size-seven deletion escape in every cubic realization. Thus every minimum extendable projection of size at most fourteen is cleanable.

We next prove a universal relative-$\mathrm{GL}(2,2)$ tensor lemma. Given an arbitrary linear functional of $L_a^{-1}L_b$ on every pair of complement components, component maps can always be chosen so that the resulting pair graph is Eulerian. A short human proof uses a three-point linear functional on the six elements of $\mathrm{GL}(2,2)$; after expanding an indicator polynomial, leaf terms vanish and nonempty cycle-union terms cancel in pairs. This implies that every charge-valid one-support-circuit boundary state is directly cleanable, with no bound on the circuit length or number of complement components. It also cleans several support circuits whenever every component is charge-balanced separately on each circuit.

At size fifteen the primary exact classifier enumerates 26,492 word orbits and 35,247,202,556 charge-valid abstract states over all eight support shapes. Of the 31,088,622,592 dirty states, 31,086,255,789 clean directly and 2,360,767 admit a strict circuit deletion. The remaining 6,036 states all have shape $7+8$. A separately written canonical audit identifies every one as a same-block one-occurrence extension of a frozen size-fourteen residual, and realization-robust inverse Kempe certificates cover all of them. The 20,004-word one-circuit row independently agrees with a zero-sum-partition recurrence; all 23,398,774,592 dirty states also clean in the completed anchor census, in agreement with the tensor proof. Therefore every minimum extendable projection of size at most fifteen is cleanable, and a counterexample to the proposed selection principle must have minimum size at least sixteen.

The support-preserving boundary method is nevertheless sharp: at size fourteen an explicit $7+7$ state is neither directly cleanable nor circuit-deletable. It has a simple bridgeless cubic realization on 18 vertices whose displayed size-fourteen projection is uncleanable under every extension. The graph's minimum projection size is nevertheless five and all four minima are cleanable, so this is a counterexample to the unrestricted boundary dichotomy, not to the minimum-projection principle or FiveCDC. The new theorem escapes only after a support-neutral Kempe recolouring.

We also delimit what the four initial exchange inequalities can prove. A K_4-e two-pole inflation preserves the support, complement components, and all two-colour boundary path pairings, while making every non-support traversal arbitrarily long. Once every support circuit uses all four affine colours, sufficiently long inflation forces all four exchange inequalities for any displayed boundary realization. An explicit 230-vertex simple bridgeless cubic example satisfies the inequalities and defeats every one-round fixed-colour Kempe multiswitch, but reaches a strict circuit deletion in two rounds. Thus the unresolved step must exploit the *dynamic* reapplication of the exchange theorem after neutral recolouring, not only its four inequalities at the initial flow.

Two further structural results sharpen that dynamic frontier. The tensor theorem directly cleans every one-circuit support, strictly subsuming the earlier two-terminal-component argument, and gives the circuitwise-balanced several-circuit corollary above. Conversely, chains of the edge-deleted $K_{3,3}$ two-pole reflect all boundary-changing Kempe moves. They transfer any hypothetical goal-free boundary orbit to one in which all four exchange inequalities hold at every reachable state. No goal-free base orbit is known, so this conditional transfer supplies neither a counterexample nor a resolution.

We perform an exact canonical census of all 41,301 connected simple cubic graphs of order 18. Among the 39,866 bridgeless graphs, exactly 179 are not Tait-colourable, and every minimum-cardinality extendable projection on all 179 is cleanable. A second exact replay uses frozen canonically generated sources: 14,009 cyclically 4-edge-connected non-Tait records through order 28, and a separate 12,892-record hard order-22 source. All 147,539 minimum extendable projections in those files are cleanable. Completeness of the source populations is inherited from their upstream generation packages; the result on the literal frozen files is checked directly. An additional exact SAT scan covers seven retained order-34 and 7,654 retained order-40 strong-snark rows. All 433,730 minimum projections in those files have size ten and are cleanable.

The universal statement suggested by this computation is recorded as an open conjecture. The missing step is to derive a colour-avoiding strict exchange from a rainbow-odd component, possibly after support-neutral recolouring. We make no claim of a FiveCDC resolution or of priority for the flow formulation. An elementary incidence-cluster construction proves that resolving the cubic conjecture would resolve the standard conjecture for arbitrary finite bridgeless multigraphs.

Contents

1 Status, conventions, and contribution boundary

A binary cycle of a graph is an Eulerian edge set; it may be empty or disconnected. A k-cycle double cover is an indexed family of k binary cycles in which every edge occurs exactly twice. Empty members may be appended, so “at most five” and “five, allowing empty members” are equivalent. The five-cycle double cover conjecture, attributed to Celmins and Preissmann, asserts that every bridgeless graph has such a cover with at most five members [1, 4].

We work with the cubic flow formulation in [3, Conjecture 3.19]. All projection and boundary theorems below apply to finite loopless cubic multigraphs unless “simple” is stated. Parallel edges are distinct edge objects. A loop would contribute twice to vertex degree; loops are excluded from those theorems to keep the cut notation literal. The finite census is restricted to simple graphs.

Proposition 1.1 (standard graph-class reduction)**.** *The standard FiveCDC assertion for finite bridgeless multigraphs, with loops and parallel edges allowed, is equivalent to its restriction to finite bridgeless loopless cubic multigraphs.*

Proof. Only the reduction to the cubic class needs proof. Discard isolated vertices. At a vertex v of incidence degree $d(v)$, make a cycle Z_v of $d(v)$ new vertices and assign the incidences at v bijectively to them. When $d(v) = 2$, Z_v is the two-edge parallel cycle. Replace every original nonloop edge by an edge joining its two assigned cluster vertices. Replace an original loop by an edge joining the two distinct cluster vertices assigned its two incidences. Every cluster vertex now has two cluster-edge incidences and one old-edge incidence, so the expansion $X(G)$ is loopless and cubic.

Every cluster edge lies on Z_v. An old edge arising from a loop has an alternative path in its cluster. For a nonloop old edge $e = uv$, bridgelessness gives a u-to-v path in $G - e$; joining consecutive attachments inside the clusters lifts it to an alternative path between the ends of e in $X(G)$. Thus $X(G)$ is bridgeless.

Let $(D_1, \ldots, D_5)$ be a five-cycle double cover of $X(G)$, and let C_i consist of the old edges in D_i, identified with the original edges of G. Sum modulo two the degree equations for D_i over all vertices of Z_v. Every cluster edge is counted twice and cancels, leaving an even number of selected original incidences at v. An original loop contributes its two incidences, exactly as required. Hence every C_i is Eulerian. Exact double coverage is inherited edge by edge, so $(C_1, \ldots, C_5)$ covers G. $\square$

Remark 1.2 (reduction scope)*.* This proposition reduces the universal graph class; it does not identify several boundary occurrences at a higher-degree vertex inside the minimum-projection argument. A separate self-contained reduction package also proves that a minimum cubic counterexample could be taken simple, non-3-edge-colourable, of girth at least five, and cyclically 4-edge-connected. These standard reductions are not used to infer the finite census populations below.

Hušek and Šámal prove that the cubic FiveCDC problem is equivalent to selecting a nowhere-zero $\mathbb{F}_2^3$-flow with an additional parity condition [3, Theorem 3.16 and Conjecture 3.19]. Their criterion is related to the prescribed-cycle formulation of Hoffmann-Ostenhof [2]. We claim no novelty or priority for either formulation.

The contribution of this note is narrower:

1. a minimum-support exchange theorem, with a complete proof;
2. a human proof through size five and an exact boundary extension through size fifteen, plus a sharp size-fourteen counterstate to the unrestricted support-preserving boundary method;
3. a universal tensor theorem that directly cleans every one-support-circuit state and every circuitwise-balanced several-circuit state;
4. four simultaneous shortest-join consequences;
5. a general inflation theorem showing that those four inequalities, at one fixed flow, do not force one-round Kempe descent;
6. a reproducible canonical order-18 census and an expanded exact replay on frozen sources through order 28; and
7. a precise conjectural step that would turn this route into a proof.

The last item is open.

2 Extendable and clean Fano projections

Let $G=(V,E)$ be a loopless cubic graph and put $K=\mathbb{F}_2^2$. Write a three-bit flow as

$$f=(h,s):E\longrightarrow \mathbb{F}_2\times K, \tag{1}$$

where $h:E\to\mathbb{F}_2$ and $s:E\to K$ are flows. We identify a binary flow with its support.

Definition 2.1. A binary cycle h, possibly zero, is *extendable* if there is a flow $s:E\to K$ for which $f=(h,s)$ is nowhere zero.

Writing $s=(p,q)$, extendability has the following elementary form:

$$E-h\subseteq p\cup q, \tag{2}$$

where p and q are binary cycles. Indeed, an edge of $E-h$ has first coordinate zero and therefore needs a nonzero low coordinate.

Fix an extension $f=(h,s)$. For $c\in K$, put

$$M_c=\{e\in h: s(e)=c\}. \tag{3}$$

Proposition 2.2. *Every M_c is a matching, and the four sets M_c partition h.*

Proof. The partition statement is immediate. If two edges incident with a cubic vertex both had flow value $(1,c)$, their sum would be zero and flow conservation would force the third incident edge to have value zero, contrary to nowhere-zeroness. □

Let W be a component of the spanning subgraph $(V,E-h)$. Every edge of $\delta(W)$ belongs to h. If

$$n_c(W)=|\delta(W)\cap M_c| \pmod 2,$$

then cut conservation for the first coordinate and the two low coordinates gives

$$\sum_{c\in K} n_c(W)=0,\qquad \sum_{c\in K} n_c(W)c=0. \tag{4}$$

The 3×4 binary matrix in (4) has one-dimensional kernel, generated by the all-ones vector. Hence the four parities $n_c(W)$ are equal.

Definition 2.3. An extension is *clean* if $n_c(W)=0$ for every component W of $(V,E-h)$, for one, and hence every, $c\in K$. A projection h is *cleanable* if it has a clean extension. A component with all four parities equal to one is *rainbow-odd.*

If $s=(p,q)$, the class M_{11} on $\delta(W)$ is $\delta(W)\cap p\cap q$. Thus cleanability is exactly the existence of binary cycles p,q satisfying (2) and

$$|\delta(W)\cap p\cap q|\equiv 0 \pmod 2 \tag{5}$$

for every component W of $(V,E-h)$. By [3, Theorem 3.16], existence of such an h is equivalent to the cubic FiveCDC assertion. In the Tait-colourable case one may take $h=\varnothing$ and a nowhere-zero K-flow.

3 The minimum-projection exchange theorem

Assume now that G is not Tait-colourable, so an extendable projection cannot be empty. Choose h of minimum cardinality among all extendable projections, and fix any extension $f = (h, s)$.

Theorem 3.1 (minimum-projection exchange)**.** *For every $c \in K$ and every binary cycle C with $C \cap M_c = \varnothing$,*

$$|C \cap h| \leq |C - h|. \tag{6}$$

Equivalently, h is a minimum-cardinality binary cycle containing M_c, simultaneously for all four $c \in K$.

Proof. Fix $c \in K$, put $t = (1, c)$, and add t to the flow value on every edge of C:

$$f'(e) = \begin{cases} f(e) + t, & e \in C, \\ f(e), & e \notin C. \end{cases} \tag{7}$$

Since C is a binary cycle, the added function is a flow. The only edges that could become zero in (7) are those whose old value was t, and these are exactly the edges of M_c. The hypothesis $C \cap M_c = \varnothing$ therefore makes f' nowhere zero.

The first coordinate of f' is $h \triangle C$. This is another extendable projection, so minimality gives

$$|h| \leq |h \triangle C| = |h| + |C - h| - |C \cap h|.$$

This is (6).

For the equivalent statement, rebase the low coordinates by replacing s with the flow $s + ch$. Its exact zero set is M_c: outside h, the low value was already nonzero, while on h it vanishes exactly when $s(e) = c$.

Let h' be any binary cycle containing M_c. The flow $(h', s + ch)$ is nowhere zero. On an edge outside h', its low value is nonzero because every zero lies in $M_c \subseteq h'$; on an edge of h', its first coordinate is one. Thus h' is extendable, and $|h| \leq |h'|$.

Conversely, if $C \cap M_c = \varnothing$, then $h \triangle C$ contains M_c. Applying the minimum-containing property to $h \triangle C$ recovers (6). ☐

3.1 Shortest joins and four-colour consequences

Let T_c be the set of endpoints of M_c. Removing M_c from a binary cycle containing M_c produces a T_c-join in $G - M_c$, and adjoining M_c reverses the correspondence. Therefore Theorem 3.1 has the following form.

Corollary 3.2. *For every $c \in K$, the edge set*

$$J_c = h - M_c$$

is a shortest T_c-join in $G - M_c$.

This is not the already-refuted principle that a cardinality-minimum exact-zero matching must extend to a FiveCDC certificate. Here the objective is the cardinality of the whole binary cycle $M_c \cup J_c$, and the same projection is optimal after all four affine rebases.

Corollary 3.3 (flow-resistance bound)**.** *Let $r_f(G)$ be the minimum number of zero edges in an $\mathbb{F}_2^2$-flow on G. Every extendable projection satisfies*

$$|h| \geq 4r_f(G). \tag{8}$$

Proof. For each c, the rebased low flow $s + ch$ has exact zero set M_c. Hence $|M_c| \geq r_f(G)$. The four M_c's partition h. $\square$

Corollary 3.4 (four-colour circuits)**.** *Every circuit component of h contains at least one edge of each of the four sets M_c.*

Proof. If a circuit component D omitted M_c, then D would be a binary cycle avoiding M_c, but $|D \cap h| = |D| > |D - h| = 0$, contrary to (6). $\square$

Theorem 3.5 (minimum projections through size five)**.** *Let G be a connected bridgeless loopless cubic graph; parallel edges are allowed. If G is not Tait-colourable and its minimum extendable projection has size at most five, then every minimum extendable projection is cleanable. If G is Tait-colourable, the zero projection is a clean minimum.*

Proof. The Tait case follows by taking a nowhere-zero K-flow as the low coordinate: its zero affine class is empty, so the equal cut parities in (4) are all even. Assume henceforth that G is non-Tait, choose a minimum extendable h, and fix an arbitrary extension $f = (h, s)$.

By Corollary 3.4, every circuit component of h meets all four M_c. Since G is loopless, $|h| \leq 5$ therefore forces h to be one ordinary circuit D of length $n = 4$ or 5; a parallel two-circuit cannot meet four classes. Write

$$v_0, e_0, v_1, e_1, \ldots, v_{n-1}, e_{n-1}, v_0, \qquad e_i = v_i v_{i+1},$$

cyclically, and put $c_i = s(e_i)$. Adjacent c_i's are distinct: otherwise the third edge at their common vertex would have full flow value zero.

Let $K_0 = G - h$. Every component of K_0 contains a vertex of D, by connectedness. Suppose the extension is dirty. For a dirty component W, membership along D toggles oddly on each of the four affine colour classes.

If $n = 4$, the four colours occur once each, so membership toggles on every e_i. The components of K_0 are exactly the two alternating vertex pairs. Indeed, separating either pair further would create a component whose cut sees exactly two affine colours oddly, contrary to (4).

If $n = 5$, rotate and rename the colours so their cyclic word is A, B, A, C, D. Put

$$z_i = 1[v_i \in W], \qquad y_i = z_i + z_{i+1}.$$

Dirtiness says

$$y_1 = y_3 = y_4 = 1, \qquad y_0 + y_2 = 1.$$

Up to complement, the solutions are $W = \{v_2, v_4\}$ and $W = \{v_1, v_4\}$. Summing the common odd cut parity over all K_0-components shows that the number of dirty components is even. The only component-compatible shore disjoint from either displayed dirty shore is its complement: exhaustive substitution in the preceding four equations also gives the clean pair $\{v_1, v_2\}$ and its complement, both of which intersect either dirty partition. Hence K_0 again has exactly two components.

Let k_i be the unique K_0-edge incident with v_i, and put

$$d_i = s(k_i) = c_{i-1} + c_i \neq 0.$$

On each of the two K_0-components, apply an independently chosen element of $\mathrm{GL}(2, 2)$ to every low flow value. This preserves internal conservation and nonzeroness. We now choose nonzero values r_i on e_i so that

$$r_{i-1} + r_i = g_i, \tag{9}$$

where g_i is the transformed value on k_i.

For $n = 4$, the boundary values on each alternating pair agree: $d_0 = d_2 = x$ and $d_1 = d_3 = y$. Map x and y, independently, to the same nonzero α. If $\{\alpha, \beta, \gamma\} = K - \{0\}$, assigning β, γ alternately to the edges of D satisfies (9).

For $n = 5$, translate the affine colour names so $A = 0$, write $B = x, C = y, D = x + y$, and normalize $x = 1, y = 2, x + y = 3$. Thus

$$d = (3, 1, 1, 2, 1).$$

The complete repairs for the two component partitions are

K_0-components on v_i	$(g_0, \ldots, g_4)$	$(r_0, \ldots, r_4)$
$\{0, 1, 3\} \mid \{2, 4\}$	$(3, 1, 2, 2, 2)$	$(2, 3, 1, 3, 1)$
$\{0, 2, 3\} \mid \{1, 4\}$	$(3, 3, 1, 2, 3)$	$(1, 2, 3, 1, 2)$.

In each row the three-terminal component uses the identity. On the two-terminal component, both original boundary values are 1, which an invertible linear map sends respectively to 2 or 3. Direct cyclic substitution verifies (9).

We have constructed a nowhere-zero K-flow on every edge of G. At a cubic vertex its three incident values are therefore the three distinct nonzero elements of K, giving a Tait colouring. This contradicts the hypothesis. Thus the arbitrary extension of h was clean, and in particular h is cleanable. $\square$

Remark 3.6. The finite case split and the two repair rows are independently replayed by `verify.py` in the bundled size-five theorem package.

3.2 A line-switch lemma and the size-six/seven boundary

We need one support-preserving fact. Identify the seven nonzero full flow values with $1, \ldots, 7$ under bitwise addition, so that

$$L = \{1, 2, 3\} = \{(0, t) : t \in K - \{0\}\}, \qquad A = \{4, 5, 6, 7\} = \{(1, c) : c \in K\}.$$

Let $\mathcal{K}$ be the components of $G - h$, and let $r \in \mathbb{F}_2^{\mathcal{K}}$ indicate the rainbow-odd components. For $t \in L$, put $N_t = f^{-1}(t)$ and $P_t = \{4, 4 + t\}$. If C is a binary cycle of $G - N_t$, define

$$\tau_t(C)_W = |C \cap \delta(W) \cap f^{-1}(P_t)| \pmod 2 \qquad (W \in \mathcal{K}). \tag{10}$$

Adding t along C is a valid line-preserving switch: avoiding N_t prevents a zero value, and (10) is its change to the defect vector.

Proposition 3.7 (combined-line span)**.**

$$r \in \sum_{t \in L} \operatorname{im} \tau_t. \tag{11}$$

Consequently, if $G - h$ has at most two components, then h is cleanable by at most one valid line-preserving switch.

Proof. Suppose (11) fails. There is a vector $y \in \mathbb{F}_2^{\mathcal{K}}$ with

$$y \cdot r = 1, \qquad y \cdot \tau_t(C) = 0 \quad (t \in L,\ C \text{ a cycle of } G - N_t).$$

Let U be the union of the components selected by y, and let $u : V(G) \to \mathbb{F}_2$ indicate U. Cycle–cut orthogonality says that

$$R_t = \delta(U) \cap f^{-1}(P_t)$$

is a cut in $G - N_t$. Choose a vertex indicator x_t for that cut, and write $q = (u, x_1, x_2, x_3)$.

Define

$$\mathcal{A}(q) = \big(u(x_2 + x_3),\ u(x_1 + x_3),\ x_1x_2 + x_1x_3 + x_2x_3\big) \in \mathbb{F}_2^3.$$

Across an edge, the only possible nonzero differences of $(u; x_1x_2x_3)$, indexed by its flow value, are

$f(e)$	1	2	3	4	5	6	7
Δq	$(0; 100)$	$(0; 010)$	$(0; 001)$	$(1; 111)$	$(1; 100)$	$(1; 010)$	$(1; 001)$.

The zero difference is also allowed in every column. Direct substitution gives the pointwise identity

$$\big(\mathcal{A}(q_v) + \mathcal{A}(q_w)\big) \cdot f(vw) = \begin{cases} 1, & f(vw) = 4 \text{ and } u(v) \neq u(w), \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

Summing (12) over all edges and regrouping at vertices makes its left side

$$\sum_v \mathcal{A}(q_v) \cdot \left(\sum_{e \ni v} f(e) \right) = 0$$

by flow conservation. The right side is $|\delta(U) \cap f^{-1}(4)| = y \cdot r = 1$, a contradiction. This proves (11).

Every vector in every image of τ_t has even Hamming weight, since an affine edge between two $G - h$ components is counted at both ends. For one component the even-weight space is zero. For two components it is one-dimensional; if $r \neq 0$, (11) forces one of the three image spaces to contain its unique nonzero vector. The corresponding single valid switch kills r. A single switch has no mixed-switch quadratic correction, completing the proof. □

Theorem 3.8 (minimum projections through size seven)**.** *Under the hypotheses of Theorem 3.5, every minimum extendable projection of size at most seven is cleanable.*

Proof. Only sizes six and seven remain. The four-colour circuit corollary forces h to be one circuit

$$v_0, e_0, v_1, e_1, \ldots, v_{n-1}, e_{n-1}, v_0, \qquad n \in \{6, 7\}.$$

Put $c_i = s(e_i)$, and encode the component of $G - h$ containing v_i by π_i. The c_i's form a proper cyclic word using all four affine colours. Conservation on every component W gives

$$\sum_{\pi_i = W} (c_{i-1} + c_i) = 0,$$

and its four affine cut parities are either all zero or all one.

The exact finite boundary classifier enumerates every such word and every set partition π. For each state it also enumerates one independent element of $\mathrm{GL}(2, 2)$ on the low flow values of every component and solves the n cycle equations. Its complete counts are

n	valid dirty pairs	failed linear repairs	failures with more than two components
6	2712	432	0
7	26880	5376	0

Every dirty state either produces a nowhere-zero K-flow on all of G, contradicting the non-Tait hypothesis, or has exactly two components of $G - h$. At size six the failed states may additionally be quotiented by the affine group on the four colours, the dihedral group of the cycle, and component relabelling; they give the three representatives

$(c_0, \ldots, c_5)$	$(\pi_0, \ldots, \pi_5)$
010123	001001
010232	001010
012013	000101

There are 144 labeled states per orbit. At size seven there is no residual normal form after the two-component theorem. The dependency-free enumerator regenerates all words, partitions, conservation tests, dirty states, and linear repairs rather than trusting the stored counts.

Proposition 3.7 makes every failed state clean by one valid switch. Hence no dirty size-six or size-seven global minimum is uncleanable. □

Remark 3.9. The exact classifications, proof of semantics, size-six representatives, and full standalone span proof are bundled in the cumulative through-seven package. This is a finite boundary theorem, not a search over sampled graphs.

3.3 Direct component repair through size ten

The preceding Tait repair required every new low value to be nonzero. For the original three-bit flow this is unnecessarily restrictive on h: an edge of h has first coordinate one and therefore remains nonzero even when its low value is zero. Exploiting this observation gives a stronger repair.

Proposition 3.10 (direct boundary repair certificate). *Let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2 \times K$-flow. Index the vertices and edges of each circuit of h cyclically, put*

$$c_i = s(e_i), \qquad d_i = c_{i-1} + c_i,$$

and let π_i denote the component W_{π_i} of $G - h$ containing the circuit vertex v_i. Suppose there are maps $L_a \in \mathrm{GL}(2, 2)$ and values $r_i \in K$ such that

$$r_{i-1} + r_i = L_{\pi_i} d_i \tag{13}$$

on every circuit, and, for every component W_a and every $c \in K$,

$$\big|\{e_i \in \delta(W_a) : r_i = c\}\big| \equiv 0 \pmod 2. \tag{14}$$

Then h is cleanable.

Proof. Apply L_a to the low value of every edge in W_a. At a vertex outside h, all three incident edges lie in the same component, so linearity preserves the flow equation. At a circuit vertex v_i, the unique edge outside h receives value $L_{\pi_i} d_i$, and (13) is exactly the missing vertex equation. Every edge outside h retains a nonzero low value because L_a is invertible. Every edge of h has full value $(1, r_i)$, which is nonzero even if $r_i = 0$. The repaired flow is therefore nowhere zero. Equation (14) is precisely the clean affine-class cut condition, component by component. □

Theorem 3.11 (minimum projections through size ten). *Let G be a connected bridgeless loopless cubic graph; parallel edges are allowed. Every cardinality-minimum extendable projection of size at most ten is cleanable.*

Proof. The zero projection is clean. For a nonzero minimum projection, Corollary 3.4 says that every circuit component uses all four affine colours and hence has length at least four. The possible length partitions through total size ten are therefore

$$\begin{gathered} 4,5,6,7,8,9,10, \\ 4+4, \quad 4+5, \quad 4+6, \quad 5+5. \end{gathered} \tag{15}$$

For a fixed shape, the exact checker enumerates every proper cyclic word on K using all four colours on every circuit, modulo the global affine action and the exact dihedral action on each circuit (including circuit interchange when the lengths agree). For every canonical word it enumerates every set partition π of the circuit vertices. It retains exactly those partitions satisfying component conservation

$$\bigoplus_{\pi_i=a} d_i = 0$$

and the four equal affine cut parities, with at least one dirty component. It then exhausts all six choices of L_a independently for every component and all four starting values independently on every circuit; equation (13) forces the remaining r_i's. Finally it tests (14) literally.

The complete counts are

shape	canonical words	valid dirty states	failed direct repairs
4	1	1	0
5	1	2	0
6	5	27	0
7	7	116	0
8	26	1386	0
9	49	8840	0
10	151	102,290	0
$4+4$	2	114	0
$4+5$	2	372	0
$4+6$	11	7798	0
$5+5$	6	4232	0

These are all shapes in (15), and every retained boundary state has a certificate of Proposition 3.10. Hence every minimum projection in the stated range is cleanable. □

Remark 3.12. The dependency-free replay is the `verify.py` program in the bundled through-ten direct-cleaning package. It regenerates the words, partitions, conservation tests, and repairs. A separately written through-nine package agrees on the overlapping range, and two independent agent runs reproduced the size-ten counts. These checks are not independent human peer review.

3.4 The first direct-repair failures and size eleven

The direct-cleaning statement in Theorem 3.11 is not universal. A second elementary certificate explains why its first failures nevertheless cannot be globally minimum.

Proposition 3.13 (circuit-deletion certificate)**.** *Let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2 \times K$-flow, and let D be a circuit component of h. Suppose component maps L_a and values r_i satisfy the vertex equations (13), and suppose $r_i \neq 0$ on every edge of D. Then $h - D$ is extendable.*

Proof. Use first coordinate one on $h - D$ and zero on D. On all circuit edges use the low values r_i, and inside each component of $G - h$ use $L_a s$. The proof of Proposition 3.10 verifies all vertex equations and nonzeroness off D. On D, the first coordinate is now zero but the assumed low value is nonzero. The result is therefore a nowhere-zero flow with first-coordinate support $h - D$. $\square$

Proposition 3.14 (fixed-map affine cleaning)**.** *Fix component maps L_a for which the circuit equations (13) are consistent, and fix one solution r^0. Let $\mathcal{D}$ be the set of circuit components of h. Every solution is of the form*

$$r_i = r_i^0 + z_D \qquad (e_i \in D),$$

with one independently chosen translation $z_D \in K$ per circuit. For a component W_a, define

$$b_a = \sum_{e_i \in \delta(W_a)} q(r_i^0), \qquad t_{aD} = \bigoplus_{e_i \in \delta(W_a) \cap D} r_i^0,$$

where $q(x_1, x_2) = x_1 x_2$, and put $B(x, y) = x_1 y_2 + x_2 y_1$. Then the repaired extension is clean exactly when

$$b_a + \sum_{D \in \mathcal{D}} B(t_{aD}, z_D) = 0 \qquad \text{for every } a. \tag{15a}$$

In particular, cleanliness for fixed component maps is an affine linear system over $\mathbb{F}_2$, with at most $2(|\mathcal{D}| - 1)$ scalar variables.

Proof. The difference of two solutions of (13) is equal on consecutive edges, and hence is constant around each circuit. Every circuit meets $\delta(W_a)$ evenly. Summing (13) at all circuit vertices in W_a also gives $\bigoplus_{e_i \in \delta(W_a)} r_i = 0$. Consequently the four parities in (14) are either all zero or all one: their total is zero, and their two K-coordinate sums are zero.

The common defect bit equals $\sum_{e_i \in \delta(W_a)} q(r_i)$. Indeed, $\mathbf{1}_{x=0} = 1 + x_1 + x_2 + q(x)$, and the constant and linear sums vanish by the preceding two parity identities. Substitute $r_i = r_i^0 + z_D$ and use

$$q(x + z) = q(x) + q(z) + B(x, z).$$

The $q(z_D)$ term occurs an even number of times on each circuit cut and cancels, leaving exactly (15a). Finally $\bigoplus_D t_{aD} = 0$, so translating every circuit by the same element changes no equation; one z_D may be fixed to zero. $\square$

Remark 3.15. This proposition is a human-checkable reduction of the translation search, not a universal solution. If its affine system is inconsistent, binary linear duality gives a short XOR witness y with $y^\mathsf{T}\Phi = 0$ and $y^\mathsf{T} b = 1$. The exact remaining issue is to choose the component maps so that either this system is consistent or one integrated circuit word omits a colour, which triggers Proposition 3.13.

Proposition 3.16 (dual-cut feasible move)**.** *If the fixed-map system (15a) is inconsistent, there is a nonempty proper union Y of components of $G - h$ such that, for every circuit D of h,*

$$\bigoplus_{e_i \in \delta(Y) \cap D} r_i^0 = 0, \tag{15b}$$

while an odd number of the circuits meet $\delta(Y)$ oddly in each of the four low colours. Moreover, for every $T \in \mathrm{GL}(2, 2)$, postcomposing all L_a with T for $W_a \subseteq Y$, and changing no other component map, preserves circuit consistency.

Proof. Let y be a dual XOR witness from the preceding remark and let Y be its selected component union. Summing t_{aD} over the selected rows cancels support edges internal to Y twice and leaves exactly the left side of (15b). Nondegeneracy of B and $y^\mathsf{T}\Phi = 0$ make this xor zero for every circuit. Each circuit crosses $\delta(Y)$ evenly, so zero colour xor forces its four cut-colour parities to be equal. The identity $y^\mathsf{T}b = 1$ says that the sum of their q-parities is odd; hence an odd number of circuits are rainbow-odd on the cut.

Finally, summing (13) over the circuit vertices in Y gives zero by (15b). Postcomposition sends this partial sum to $T(0) = 0$; the complementary partial sum is also zero because the original circuit total was zero. Thus every circuit equation remains consistent. ☐

Remark 3.17. The proposition turns every failed translation system into six coordinated feasible map choices. A universal proof still needs a well-founded progress statement: one of these choices must clean, make some circuit omit a colour, or decrease an invariant. No such invariant is proved here.

Theorem 3.18 (minimum projections through size eleven)**.** *Under the hypotheses of Theorem 3.11, every cardinality-minimum extendable projection of size at most eleven is cleanable.*

Proof. Only total size eleven remains. The four-colour circuit condition gives the three shapes

$$11, \qquad 4+7, \qquad 5+6. \tag{16}$$

The same exact boundary enumeration used above gives

shape	canonical words	charge-valid	dirty	failed direct repairs
11	345	1,183,710	919,764	0
$4+7$	17	57,500	46,786	0
$5+6$	26	87,884	71,170	14

The 14 literal failures form 11 orbits under the full affine, dihedral, and component-renaming action. For each failure, the checker exhausts the same component maps and independent circuit starts and finds a certificate satisfying (13) in which every low value on the six-circuit is nonzero. Proposition 3.13 then produces an extendable projection supported only on the five-circuit. Its size is five, contradicting the assumed global minimality of the size-eleven projection. Hence every globally minimum size-eleven state is directly cleanable. ☐

Remark 3.19 (sharp direct-repair countermodel). The 14 failed boundary states are realizable. Replacing each two-vertex component block by an edge and each three-vertex block by a claw yields, in all 14 cases, the same simple bridgeless non-Tait graph on 12 vertices, with graph6 encoding

`Kt?G?DIPOqCo`.

It is the triangle-expanded Petersen graph. Exact cycle-space enumeration shows that the displayed size-eleven 5+6 projection is extendable but uncleanable, while the minimum extendable-projection size is five and all six size-five minima are cleanable. Thus it refutes the direct-repair strengthening, not FiveCDC or the minimum-projection conjecture. The cumulative checker and all 14 certificates are in the bundled through-eleven package.

3.5 Clean or delete and Kempe escape through size fourteen

Propositions 3.10 and 3.13 give a useful dichotomy: it is enough to find component maps and circuit starts that either make every affine cut parity even, or make one whole support circuit low-valued and nowhere zero. The first branch cleans the projection; the second strictly shortens it and is therefore impossible for a global minimum.

Theorem 3.20 (minimum projections through size twelve)**.** *Under the hypotheses of Theorem 3.11, every cardinality-minimum extendable projection of size at most twelve is cleanable.*

Proof. Only total size twelve remains. The complete circuit-length list is

$$12, \qquad 4+8, \qquad 5+7, \qquad 6+6, \qquad 4+4+4. \tag{17}$$

For each shape, the exact classifier enumerates the same word orbits and charge-valid component partitions as before. It first exhausts every component map and relative circuit start for direct cleanliness. If that fails, it tests whether the integrated word on a circuit omits a colour. Translating that circuit by the omitted colour makes all of its low values nonzero, giving the hypothesis of Proposition 3.13. The mutually exclusive counts are

shape	words	dirty	direct clean	delete
12	1014	11,465,978	11,465,978	0
$4+8$	66	773,721	773,717	4
$5+7$	64	738,969	738,763	206
$6+6$	66	774,196	774,014	182
$4+4+4$	3	35,960	35,960	0

There are zero states in neither branch. The 392 deletion states have literal component maps, integrated circuit words, and omitted-colour certificates in the frozen output. Since a globally minimum projection cannot take the deletion branch, every size-twelve minimum is cleanable. ☐

Remark 3.21. The primary C++ classifier and certificate parser are bundled in the through-twelve clean-or-delete package. A separately written C++ exact cover auditor independently regenerates the word orbits, valid component blocks, map search, clean decisions, and all 392 deletion decisions. The two implementations agree on every table entry. This is independent agent replication, not human peer review.

Theorem 3.22 (minimum projections through size thirteen)**.** *Under the hypotheses of Theorem 3.11, every cardinality-minimum extendable projection of size at most thirteen is cleanable.*

Proof. Only total size thirteen remains. Since every circuit component contains all four colours, the complete shape list is

$$13, \qquad 4+9, \qquad 5+8, \qquad 6+7, \qquad 4+4+5. \tag{18}$$

The exact classifier applies the same clean-or-delete test as in Theorem 3.20. Its complete counts are

shape	words	charge-valid	dirty	direct clean	delete
13	2583	155,381,679	129,684,490	129,684,490	0
$4+9$	130	7,753,918	6,657,654	6,657,594	60
$5+8$	203	12,144,485	10,404,652	10,401,428	3224
$6+7$	242	14,480,258	12,415,026	12,410,636	4390
$4+4+5$	4	238,522	208,144	208,144	0

Thus 159,369,966 dirty states split into 159,362,292 direct repairs and 7,674 strict circuit deletions, with no residual state outside those branches. Proposition 3.13 excludes every deletion state from global minimality, proving the claim. □

Remark 3.23. The primary size-thirteen package freezes one literal map, integrated word, deleted circuit, and omitted colour for every deletion state. A separately written valid-block/exact-cover implementation regenerates the word orbits, component partitions, map search, affine cleaning decisions, and deletion decisions. The implementations agree on every table entry. This is exact independent agent replication, not peer review.

Proposition 3.24 (sharp failure of unrestricted clean or delete)**.** *The clean-or-delete boundary statement used in Theorems 3.20 and 3.22 is false at total support size fourteen. A counterstate has two seven-circuits with*

$$\begin{aligned}(c_i) &= \mathtt{0101023} \mid \mathtt{0101232},\\ (\pi_i) &= \mathtt{01234444413024}.\end{aligned} \tag{19}$$

Every circuit uses all four colours and every component charge is zero, but no component-map tuple and relative circuit translation cleans the state or makes either circuit low-valued and nowhere zero.

Proof. The derivative word is `3111121 | 2111311`, from which the five zero block charges and dirtiness are checked directly. Normalize the map of block zero to the identity, and put

$$u_i = L_i(1)\ (i = 1,2,3), \qquad a = L_4(1), \qquad b = L_4(2).$$

All five displayed elements are nonzero and $a \neq b$. The transformed derivatives on the two circuits are

$$(3, u_1, u_2, u_3, a, b, a), \qquad (b, a, u_1, u_3, 3, u_2, a).$$

Their two integrability equations coincide and reduce to

$$b = 3 + u_1 + u_2 + u_3. \tag{20}$$

Zero-start integration gives

$$\begin{aligned}r^A &= (3, 3+u_1, 3+u_1+u_2, b, a+b, b, a, 0),\\ r^B &= (b, a+b, 3+u_2+u_3+a, 3+u_2+a, u_2+a, a, 0).\end{aligned}$$

Both rows contain $\{0, a, b, a+b\} = K$, so deletion is impossible.

Let z translate the second circuit. Cleanliness at blocks $0, 1, 2, 3$ respectively requires z to lie in

$$\begin{aligned}&u_2 + a + \{0, 3\}, && 3 + a + b + \{0, u_1\},\\ &3 + u_1 + u_2 + a + \{0, u_2\}, && u_1 + u_3 + a + \{0, u_3\}.\end{aligned} \tag{21}$$

If $z = u_2 + a + 3$, the third and fourth lines force $u_1 = u_2$ and $u_3 = 3$; then (20) gives $b = 0$. If $z = u_2 + a$, the second and fourth lines force $u_3 = u_1$ and $u_2 = u_1$; the third then forces $u_1 = 3$, and again (20) gives $b = 0$. Both conclusions contradict invertibility of L_4. Thus no clean translation exists. □

Remark 3.25 (realization and scope). The state (19) is realized by a simple connected bridgeless non-Tait cubic graph on 18 vertices, with canonical graph6 encoding

`Qs???SC@GS@_CDOoC@@@?O?CO?g.`

The graph is nonplanar; the package displays a $K_{3,3}$ subdivision. Exact enumeration of its 2^{10}-element cycle space finds 15,360 ordered extensions of the displayed size-fourteen projection and no clean one. Nevertheless, the graph has minimum extendable-projection size five, exactly four minimum supports, and all four are cleanable. Thus Proposition 3.24 refutes the unrestricted boundary lemma, not Conjecture 4.1 or FiveCDC. The complete certificate and graph audit are in the bundled size-fourteen dichotomy-counterstate package; it also independently replays all 1,296 normalized map tuples and freezes the 40 effective cases. A separately written package exhausts all 6^5 full map assignments and independently reconstructs the graph, cycle space, extension counts, and four minima.

Theorem 3.26 (minimum projections through size fourteen)**.** *Under the hypotheses of Theorem 3.11, every cardinality-minimum extendable projection of size at most fourteen is cleanable.*

Proof. Only total size fourteen remains. Since every circuit component contains all four affine colours, the complete shape list is

$$\begin{gathered} 14, \quad 4+10, \quad 5+9, \quad 6+8, \quad 7+7, \\ 4+4+6, \quad 4+5+5. \end{gathered} \tag{22}$$

The primary exact classifier applies the same component-map, relative-start, cleaning, and deletion tests as above. Its mutually exclusive counts are

shape	words	dirty	direct clean	delete	residual
14	7382	1,748,842,896	1,748,842,896	0	0
$4+10$	416	100,736,602	100,735,166	1436	0
$5+9$	505	121,073,174	121,032,843	40,331	0
$6+8$	809	195,911,662	195,844,374	67,288	0
$7+7$	333	80,104,020	80,066,557	37,239	224
$4+4+6$	22	5,408,156	5,408,156	0	0
$4+5+5$	14	3,401,666	3,401,596	70	0

(23)

The seven rows comprise 9,481 word orbits and 2,616,134,989 charge-valid states. The 2,255,478,176 dirty states split into 2,255,331,588 direct repairs, 146,364 strict circuit deletions, and 224 residuals. The deletion rows contradict global minimality by Proposition 3.13. Every residual has shape $7+7$, with 46 states of component-size profile $6+2+2+2+2$ and 178 of profile $4+4+2+2+2$.

It remains to exclude those 224 states. Fix two distinct nonzero low colours x, y, put $\Delta = x+y$, and consider one component W of $G-h$. Retain the edges of W whose low value is x or y. There is a linear functional $\ell : K \to \mathbb{F}_2$ satisfying

$$\ell(x) = \ell(y) = 1, \qquad \ell(\Delta) = 0. \tag{24}$$

Applying ℓ to the low-flow equation shows that every vertex of W away from h has even retained degree. At the split boundary occurrence i, the unique complement edge has low value d_i. Consequently the odd boundary vertices are exactly the occurrences with $d_i \in \{x, y\}$, and the retained subgraph pairs them by paths.

Interchange x and y on one such path. Each internal flow equation is unchanged, every complement value remains nonzero, and both endpoint derivatives change by Δ. Since both endpoints belong to the same complement component, its boundary charge is unchanged. If the endpoints lie on the same support circuit, its two changes cancel. If they lie on different support circuits, the certificate names another component whose selected two-colour subgraph has exactly two odd boundary occurrences, one on each circuit. Its forced path supplies a second switch by Δ, restoring both circuit closure equations.

The pairing of terminals depends on the unknown realization inside W. For each residual the finite certificate therefore stores a set S of successful endpoint pairs such that

$$M \cap S \neq \varnothing \quad \text{for every perfect matching } M \text{ of the terminals.} \tag{25}$$

There are three possible perfect matchings on four terminals and fifteen on six. For every pair in S, the certificate gives the primary and auxiliary endpoints, all component maps, both integrated circuit words, and a colour omitted from one circuit. Translating by that omitted colour and deleting the circuit produces an extendable projection of size seven.

The exhaustive builder verifies all normalized map choices and produces 724 literal deletion rows. A separately structured checker verifies (25), every path-switch boundary update, every component map and circuit integration, and every omitted-colour deletion. Hence every possible internal path pairing yields a strict deletion. No residual is globally minimum, which proves the theorem. □

Remark 3.27 (scope and independent replication). The path argument uses the split-occurrence boundary model, automatic in a cubic graph: every circuit occurrence is a distinct vertex with one incident complement edge carrying d_i. If several occurrences are identified at one higher-degree vertex, only their xor is visible to the complement two-colour subgraph. Thus the path theorem itself is not a higher-degree boundary theorem. Proposition 1.1 instead replaces the graph by a new cubic expansion and pushes a completed cover back.

A separately written $7+7$ census regenerates all 333 word orbits, 91,481,505 charge-valid states, 80,104,020 dirty states, and the same 224 residual rows using a valid-block exact-cover implementation. Its sorted failure-set digest agrees with the primary census. This is independent agent replication, not independent human peer review. Only the $7+7$ branch has an independently implemented full state census. For the other six shapes—14, $4+10$, $5+9$, $6+8$, $4+4+6$, and $4+5+5$—the word-orbit counts, charge-valid partition counts, and all direct-clean/delete decisions currently rest on the primary classifier.

3.6 A universal tensor repair and the size-fifteen frontier

The one-circuit rows in the preceding finite tables are instances of a general algebraic theorem.

Theorem 3.28 (relative-GL(2,2) tensor lemma)**.** *Put $\mathcal{G} = \mathrm{GL}(2,2)$. Let V be finite. Orient each unordered pair $\{a,b\} \subseteq V$ once, and assign it an arbitrary linear functional*

$$\phi_{ab} : \mathrm{Mat}_2(\mathbb{F}_2) \longrightarrow \mathbb{F}_2.$$

There are maps $L_a \in \mathcal{G}$ such that, for

$$w_{ab} = \phi_{ab}(L_a^{-1} L_b), \qquad Q_a = \sum_{b \neq a} w_{ab}, \tag{26}$$

every Q_a is zero.

Proof. Regard functions $\mathcal{G} \to \mathbb{F}_2$ as a six-dimensional vector space. Identify $0 = (0,0)$, $1 = (1,0)$, $2 = (0,1)$, and $3 = (1,1)$. Write a map as the string of its images of $0,1,2,3$, and define

$$\lambda(f) = f(\mathtt{0132}) + f(\mathtt{0213}) + f(\mathtt{0321}).$$

Then

$$\lambda(1) = 1, \qquad \lambda(M \mapsto M_{ij}) = 0 \quad (1 \le i, j \le 2). \tag{26a}$$

For the second assertion, in column-major order the flattened matrices of the three displayed maps are

$$(1,0,1,1), \quad (0,1,1,0), \quad (1,1,0,1),$$

whose sum is zero.

For a fixed tensor instance, let

$$F((L_a)_{a\in V}) = \prod_{a\in V}(1+Q_a).$$

This is the indicator of the desired assignments. Apply $\Lambda = \lambda^{\otimes V}$ after expanding F. An expansion choice selects at most one incident edge at each vertex. Because the edge functions are Boolean, $w_e^2 = w_e$, so terms may be grouped by the simple graph H of distinct selected edges.

If a nonempty H has a leaf, its grouped monomial depends on the map at that leaf through one linear coordinate function. Multiplication by a fixed matrix is linear, and on $\mathcal{G}$

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix}^{-1} = \begin{pmatrix} s & q \\ r & p \end{pmatrix}.$$

Thus (26a) kills the monomial, regardless of which orientation was fixed for the leaf edge.

Suppose instead that H has no leaf. If S is the set of vertices that selected an edge, then

$$|E(H)| \le |S| \le |V(H)| \le |E(H)|.$$

The last inequality is the minimum-degree-two bound, with $V(H)$ containing only nonisolated vertices. Equality holds throughout. Consequently H is a disjoint union of cycles, every vertex selects one edge, and every edge is selected exactly once. Each cycle has two such selections, its two cyclic orientations. A nonempty union of cycles therefore has coefficient $2^{c(H)} = 0$ in $\mathbb{F}_2$.

All nonconstant grouped terms are killed or cancel. The constant term gives

$$\Lambda(F) = \lambda(1)^{|V|} = 1.$$

Hence F is not the zero function, proving the theorem. □

Corollary 3.29 (universal one-circuit direct cleaning)**.** *Every charge-valid abstract boundary state whose support is one circuit has a direct repair of Proposition 3.10. There is no bound on the circuit length, number of complement components, or component degrees in the abstract state.*

Proof. Use the notation of Proposition 3.10, and assume the single support circuit is cut open in a fixed place. For each component a, conservation says

$$\bigoplus_{\pi_i = a} d_i = 0. \tag{26b}$$

Choose local maps L_a and put $z_i = L_{\pi_i} d_i$. For distinct components a, b, define

$$w_{ab} = \sum_{\substack{j<i \\ \pi_j = b,\ \pi_i = a}} B(z_j, z_i), \tag{26c}$$

where B is the nondegenerate alternating form on $\mathbb{F}_2^2$. The opposite ordering has the same value, since the sum of the two orderings is

$$B\left(\sum_{\pi_j=b} z_j, \sum_{\pi_i=a} z_i\right) = 0$$

by (26b). Thus w_{ab} is symmetric and independent of the cut. For the latter assertion explicitly, moving one occurrence (a, z) across the cut changes its pair bit with b by $B(z, \sum_{\pi_i=b} z_i) = 0$; the case of a b-occurrence is the same. For the orientation fixed on $\{a, b\}$, preservation of B gives

$$B(L_b d_j, L_a d_i) = B(L_a^{-1} L_b d_j, d_i).$$

Summing these terms defines a linear functional of the four matrix entries of $L_a^{-1} L_b$, exactly as in Theorem 3.28. Reversing the chosen pair orientation is harmless because matrix inversion is linear on $\mathcal{G}$.

It remains to identify the obstruction. Integrate the z_i's around the circuit, with an arbitrary base value, to obtain the r_i's in (13). Put $q(x_1, x_2) = x_1 x_2$; its polarization identity is

$$q(x+y) + q(x) + q(y) = B(x, y). \tag{26d}$$

The common parity of the four low colours at component a is the parity of colour 11. At the boundary occurrence i, both support incidences e_{i-1}, e_i must be counted; an edge whose two ends belong to the same component is thereby counted twice and cancels. Hence this parity is

$$P_a = \sum_{\pi_i=a} \big(q(r_{i-1}) + q(r_i)\big).$$

Since $r_i = r_{i-1} + z_i$, (26d) gives

$$P_a = \sum_{\pi_i=a} q(z_i) + \sum_{\pi_i=a} B(r_{i-1}, z_i). \tag{26e}$$

After cutting the circuit, write $r_{i-1} = r_* + \sum_{j<i} z_j$. The base term in (26e) is $B(r_*, \sum_{\pi_i=a} z_i) = 0$ by (26b), so

$$P_a = \sum_{\pi_i=a} q(z_i) + \sum_{\substack{j<i \\ \pi_i=a}} B(z_j, z_i).$$

Separate the last sum according to whether $\pi_j = a$. Applying q to the zero block sum in (26b) gives

$$0 = q\left(\sum_{\pi_i=a} z_i\right) = \sum_{\pi_i=a} q(z_i) + \sum_{\substack{j<i \\ \pi_j=\pi_i=a}} B(z_j, z_i).$$

Thus the $q(z_i)$ terms and all same-block pairs cancel. The remaining cross-component terms are exactly

$$Q_a = \sum_{b \neq a} w_{ab}.$$

Theorem 3.28 chooses the maps so that all these common parity bits vanish. Equations (13) and (14) then hold, so the state directly cleans. $\square$

Corollary 3.30 (circuitwise-balanced several-circuit repair)**.** *Suppose a boundary state has support circuits $D_1, \dots, D_t$ and satisfies*

$$\bigoplus_{\substack{\pi_i = a \\ i \text{ on } D_j}} d_i = 0 \qquad \textit{for every component } a \textit{ and circuit } D_j. \tag{26f}$$

Then it is directly cleanable.

Proof. Construct (26c) separately on every circuit. For each unordered pair, sum its per-circuit linear functionals. The total obstruction at a component is the sum of its per-circuit obstructions, hence is again the degree parity Q_a of one tensor instance. Apply Theorem 3.28 once to obtain a common map L_a for every circuit. Condition (26f) makes each circuit integrable, since

$$\bigoplus_{i \text{ on } D_j} L_{\pi_i} d_i = \bigoplus_a L_a \left(\bigoplus_{\substack{\pi_i = a \\ i \text{ on } D_j}} d_i \right) = 0.$$

Integrate the circuits independently. □

Remark 3.31. Condition (26f) is stronger than ordinary component conservation, which requires only the xor over all support circuits to vanish. Thus Corollary 3.30 is a structural subclass theorem, not a replacement for the multi-circuit census below.

Theorem 3.32 (minimum projections through size fifteen)**.** *Under the hypotheses of Theorem 3.11, every cardinality-minimum extendable projection of size at most fifteen is cleanable.*

Proof. Only total size fifteen remains. The complete circuit-length list is

$$\begin{gathered} 15, \quad 4+11, \quad 5+10, \quad 6+9, \quad 7+8, \\ 4+4+7, \quad 4+5+6, \quad 5+5+5. \end{gathered}$$

The primary exact classifier applies the same canonical word generation, charge-valid set-partition enumeration, component maps, relative circuit starts, direct-cleaning test, and strict-deletion test as above. Its mutually exclusive counts are:

shape	words	charge-valid	dirty	direct clean	delete	residual
$4+4+7$	34	45,252,514	40,901,504	40,901,444	60	0
$4+5+6$	72	94,943,832	85,691,028	85,687,790	3,238	0
$5+5+5$	20	26,498,996	23,865,564	23,865,124	440	0
$4+11$	985	1,305,883,765	1,166,954,048	1,166,933,276	20,772	0
$5+10$	1,489	1,975,366,285	1,762,347,048	1,761,806,487	540,561	0
$6+9$	1,924	2,555,260,324	2,281,097,772	2,280,299,726	798,046	0
$7+8$	1,964	2,609,251,412	2,328,991,036	2,327,987,350	997,650	6,036
15	20,004	26,634,745,428	23,398,774,592	23,398,774,592	0	0
total	26,492	35,247,202,556	31,088,622,592	31,086,255,789	2,360,767	6,036

Corollary 3.29 proves the 15 row without enumeration. As an exact regression, sixteen completed anchor/Laplacian shards also test all 23,398,774,592 dirty one-circuit states and find zero failures. A separately implemented zero-sum-partition recurrence reproduces the one-circuit charge-valid, initially clean, and dirty counts.

For the other seven shapes, every deletion row contradicts global minimality by Proposition 3.13. Only the 6,036 7+8 residual rows remain. They are handled by a finite, realization-independent lift from the size-fourteen Kempe certificates. Delete one removable repeated-colour occurrence whose two adjacent derivative occurrences lie in the same complement block. If the old derivatives are a, b, the shorter boundary replaces them by $a + b$; charge conservation is unchanged and no two blocks are merged.

For a fixed two-colour Kempe pair, terminal status is a linear functional of the derivative. Every perfect matching of the old terminals collapses to one on the shorter boundary. If both a, b are terminals and are matched to p, q, the collapsed matching has the artificial pair pq; switching its two actual old paths has the same boundary action as the short switch, apart from an explicitly tracked change of the reinserted support value. The other cases merely delete the pair ab, rename one terminal as $a + b$, or leave the matching unchanged. Thus a matching-robust short certificate lifts after finitely checking the reinserted value, circuit closure, and final omitted-colour deletion.

A separately written canonical induction checker verifies every one of the 6,036 emitted residual rows. Each has between three and five same-block smoothing positions, and at least one of them smooths canonically to a frozen size-fourteen residual. Thus each row is canonically equivalent to a member of the 6,180-state inverse-extension family. A literal checker quantifies over every old perfect matching and verifies a successful one- or two-path lift for all 6,180 states; a redundant direct checker verifies the same realization-robust game on all 6,036 emitted rows. Hence every cubic realization of every size-fifteen residual has a strict deletion escape. No residual can be globally minimum, proving the theorem. □

Remark 3.33 (size-fifteen trust boundary). The tensor proof is human-checkable and independent of the one-circuit census. The zero-sum recurrence independently checks its partition totals; the anchor parser checks shard completeness and zero failures but imports the primary word/partition generator. The multi-circuit word-orbit counts, charge-valid partition counts, and clean/delete decisions rest on the primary classifier. The residual induction checker is independently written, but takes the 6,036 primary residual rows as input. It finds 5,200 fully canonical residual classes, with digest

`f2c0c2b53322c72f00672e3f9c7d807aeb26a6ba771357a673c84c2fdaf96206`.

All source, shard outputs, literal residuals, frozen totals, and hashes are in the bundled size-fifteen exact-frontier, induction-audit, anchor-audit, and one-circuit tensor packages. These computational claims are exact agent-produced certificates, not independent human peer review.

Corollary 3.34 (rainbow shortcut)**.** *Every binary cycle C satisfying $|C \cap h| > |C - h|$ meets all four sets M_c.*

Proof. If C omitted one class M_c, it would contradict (6). □

4 The exact missing step

Theorem 3.1 suggests the following selection principle.

Conjecture 4.1 (minimum extendable projection)**.** Every bridgeless cubic graph has a minimum-cardinality extendable projection that is cleanable.

The stronger statement that every minimum projection is cleanable also survives the finite census in Section 5, but it is not needed. Conjecture 4.1, together with the Hušek–Šámal equivalence and Proposition 1.1, would imply the standard FiveCDC conjecture for every finite bridgeless multigraph.

Proposition 3.24 shows that componentwise maps, circuit translations, and whole-circuit deletion do not by themselves prove this conjecture. A valid boundary can defeat every such move even when its projection is uncleanable under all extensions. What excludes the example at size fourteen is additional realization structure: the two-colour subgraphs inside complement components supply Kempe paths, and Theorem 3.26 uses them to reach a strict deletion. For larger supports no universal choice of such paths and subsequent maps is known. A remaining argument may require a general Kempe descent, the full four-colour exchange inequalities, or both.

Suppose a minimum projection h is not cleanable. Every extension then has a rainbow-odd component W of $G - h$, so all four M_c's cross $\delta(W)$ oddly. A direct proof of Conjecture 4.1 would need to derive either

1. a binary cycle C avoiding some M_c and satisfying $|C \cap h| > |C - h|$, contradicting Theorem 3.1; or

2. a sequence of support-neutral low-coordinate recolourings followed by such a strict exchange.

The second possibility cannot currently be discarded. Small exact examples have four disjoint colour classes meeting every immediately component-reducing cycle, while a support-preserving recolouring escapes the trap. Other examples show that a fixed nowhere-zero Fano flow can have all seven of its functional projections dirty. These are proof-strategy obstructions, not FiveCDC counterexamples. They rule out an argument that fixes an arbitrary initial flow or assumes one elementary switch always suffices.

The unresolved human proof obligation is precise: from a rainbow-odd component of an *uncleanable globally minimum-support projection*, derive a negative colour-avoiding cycle after allowable neutral recolouring, using the four simultaneous shortest-T_c-join inequalities of Corollary 3.2. No proof of this statement is given here. By Theorem 3.32, any counterexample to this selection principle has minimum support at least sixteen.

5 Exact finite census

5.1 Canonical order-18 method

The accompanying checker uses canonical generation rather than sampling. For order 18, the command

```
geng -cq -d3 -D3 18
```

generates every connected simple cubic graph once. The checker then:

1. tests bridgelessness by deleting every edge and checking connectivity;

2. tests Tait colourability by exhaustive backtracking;

3. enumerates the entire binary cycle space;

4. processes nonzero projections h by increasing cardinality;

5. quotients pairs of cycles (p, q) by their exact semantic state $(p \cup q, p \cap q)$;

6. tests extendability by $E - h \subseteq p \cup q$; and

7. computes the components of $G - h$ and tests (5) on every component.

For a connected cubic graph of order 18, the binary cycle-space dimension is

$$|E| - |V| + 1 = 27 - 18 + 1 = 10.$$

Thus all $2^{10} = 1024$ binary cycles are enumerated. Equal union/intersection pairs have exactly the same coverage and component parity behaviour, so the semantic quotient loses no information. No snark or cyclic-connectivity reduction is used.

5.2 Results and reproducibility

class	count
connected simple cubic graphs	41,301
bridgeless graphs	39,866
bridgeless non-Tait graphs	179

Table 1: Complete order-18 population.

Every minimum-cardinality extendable projection of every one of the 179 non-Tait graphs is cleanable. The minimum-size histogram is

$$5 : 166, \qquad 6 : 11, \qquad 7 : 1, \qquad 8 : 1.$$

The checker hashes, in canonical graph order, the graph6 string, minimum size, number of minimum extendable projections, and number of cleanable minimum projections. The resulting SHA-256 is

`82b01cc2f01d4f50f7745f4bdb1b3dc46ade798d452a9d4f11d4a91d738a7834`.

From the project root, run:

```
python3 scratch/check_husek_samal_minimum_projection_frontier.py \
  --order 18
```

The checker requires Python, the sibling cycle-space module `scratch/search_fano_all_bad_projection_subspaces.py`, and nauty `geng` on `PATH`. It has no SAT solver, network, or non-standard Python-package dependency. The relevant source hashes are

frontier checker:
`37e972782863c7a9506f6d339159ddaca3865363107704f86365c1a9b77a30c3`
cycle-space module:
`7e5ea2599e7533f041ea88de5c6f5c7a3bf542cedc33d78728439d43ce8861e8`

This is a finite theorem. It is evidence for Conjecture 4.1, not a proof of it.

5.3 Expanded frozen-source replay

A separate C++ classifier implements the same extendability and cleanability semantics by Gaussian elimination over $\mathbb{F}_2$. It replays two previously frozen graph6 populations:

1. 14,009 cyclically 4-edge-connected non-Tait records of even orders 10 through 28; and
2. 12,892 bridgeless non-Tait rows from a broader hard order-22 source.

frozen source	graphs	minimum projections	uncleanable
cyclically 4, orders 10–28	14,009	92,754	0
hard order 22	12,892	54,785	0
total	26,901	147,539	0

Table 2: Exact minimum-extendable-projection replay on frozen sources.

The exact results are:

For the cyclically 4 source the minimum-size profile is

$$5 : 14007, \qquad 6 : 1, \qquad 9 : 1.$$

For the hard order-22 source it is

$$5 : 11776, \quad 6 : 1025, \quad 7 : 55, \quad 8 : 20, \quad 9 : 6, \quad 10 : 10.$$

These counts refer to graphs; the third column of the table counts all minimum projections on those graphs.

From the project root, run:

```
cd scratch/minimum-projection-census-through28-20260729
python3 verify_summary.py
```

The replay compiles the classifier, verifies every input digest and every emitted graph6 row, and checks all aggregates above. Its checksum ledger is the `SHA256SUMS` file in that package. The unconditional claim here concerns the literal frozen files. Their identification with complete graph classes uses the documented upstream canonical-generation and filtering packages; this replay does not independently regenerate those populations.

5.4 Retained strong snarks of orders 34 and 40

A separate exact scan covers the literal retained strong-snark files at orders 34 and 40: Every row

order	graph6 rows	minimum projections	uncleanable
34	7	371	0
40	7,654	433,359	0
total	7,661	433,730	0

Table 3: Exact minimum-projection scan on the frozen strong-snark rows.

has minimum extendable-projection size ten. An ordinary-CNF solver encodes the two low binary cycles, coverage on $E - h$, product variables for $p_e \wedge q_e$, and an XOR chain imposing even product parity on every component cut. Every even projection through weight twelve is enumerated as a vertex-disjoint union of circuits, which is exhaustive for a cubic graph. All rows reach their first extendable level at weight ten.

The order-34 output and the first five order-40 rows were reproduced independently by the linear-algebra scanner used in the earlier census. The full order-40 claim uses the exact SAT implementation. The concatenated result digest is `becb7e4b648c000ec874509bccdf389df24f6f0d82bce2da49ba10d5b10d38e3`.

Inputs, frozen output, both scanner implementations, and the replay script are under `scratch/`, in the subdirectory `minimum-projection-known-strong-snarks-20260729/`. As before, the unconditional statement concerns the literal bundled rows; completeness of the named graph populations is inherited from upstream provenance.

5.5 A certified 130-vertex parameter-separation test

The retained 130-vertex exchange graph gives a substantially larger exact stress test. A three-coordinate CNF uses binary-cycle variables (h_e, p_e, q_e), the three parity equations at every vertex, the nowhere-zero clauses $h_e \vee p_e \vee q_e$, and a sequential cardinality counter. Its certified results are

$$\min |h| = 42, \qquad \#\{h : |h| = 42,\ h \text{ extendable}\} = 11264,$$

and every one of those 11,264 supports is cleanable. The size-$\leq$ 41 CNF has 7,760 variables and 16,218 clauses. A second CNF blocks every listed size-42 support and proves enumeration completeness. Both UNSAT LRATs are accepted by C `lrat-check` and verified CakeML `cake_lpr`; every positive clean lift is also checked directly.

On the same graph the different parameter

$$\rho_3(G) = \min_{f,\ a \neq 0} |f^{-1}(a)|$$

equals five, while no size-five value class packs two boundary joins. Thus minimizing one nonzero Fano value class and minimizing an entire coordinate projection are not interchangeable. The former selection principle fails on this graph; the latter passes all 11,264 minimum cases. To replay the package, run:

```
cd search/minimum-projection-n130-20260729
python3 verify.py
```

6 Discussion

The exchange theorem converts global support minimality into four ordinary geodesic conditions: one shortest join for each affine value class. Theorems 3.5, 3.8, 3.11, 3.18, 3.20, and 3.22, together with Theorems 3.26 and 3.32 close every possible nonzero support size through fifteen: a counterexample to Conjecture 4.1 must have minimum projection size at least sixteen. The useful feature is simultaneity. A cycle that would shorten the projection is allowed to be blocked by one colour, but Corollary 3.34 forces it to be blocked by all four. Likewise, disconnected projections are severely constrained because every circuit component must contain every colour. The size-fourteen state (19) shows that this numerical frontier is sharp for the support-preserving clean-or-delete boundary method. Theorem 3.26 also shows exactly what that abstract state forgets: two-colour subgraphs inside a cubic complement pair the boundary occurrences by paths, and a matching-robust Kempe switch reaches a strict deletion. Corollary 3.29 removes one entire unbounded branch: every one-circuit boundary directly cleans. The remaining open frontier therefore begins with several circuits lacking the separate per-circuit balance of Corollary 3.30; larger such states need not admit the same finite strategy.

Proposition 6.1 (static exchange has no further local boundary force)**.** *Let G be a finite connected simple bridgeless cubic graph, let H be a 2-regular binary cycle, and let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2^3$-flow with* $\operatorname{supp}(h) = H$*. Suppose every circuit component of H contains every value of $s|_H : E(H) \to \mathbb{F}_2^2$. There are a simple connected bridgeless cubic graph G' and a nowhere-zero flow $f' = (h', s')$ on G', with* $\operatorname{supp}(h') = H$ *and $s'|_H = s|_H$, carrying the same support circuits, complement-component partition, and two-colour boundary path pairings, for which*

$$C \cap M_c = \varnothing \quad \Longrightarrow \quad |C \cap H| \le |C - H| \tag{27}$$

for every $c \in \mathbb{F}_2^2$ and every binary cycle C of G', where $M_c = \{e \in H : s'(e) = c\}$.

Proof. Let $e = uv \notin H$ have nonzero low colour $t = s(e)$, and let $\{t, x, y\} = \mathbb{F}_2^2 - \{0\}$. Delete e, add vertices a, b, p, q, and add

$$ua, bv, pq \text{ of colour } t, \qquad ap, bq \text{ of colour } x, \qquad aq, bp \text{ of colour } y. \tag{28}$$

Give every new edge first coordinate zero and retain f on every unreplaced edge. Every new vertex sees the three nonzero low colours, so both flow equations are preserved and the resulting flow is nowhere zero. For either pair $\{t, x\}$ or $\{t, y\}$, the bichromatic path joins the old endpoints u, v; the $\{x, y\}$ component is internal. Hence the substitution preserves every original two-colour boundary pairing. Every u-to-v traversal through the two-pole, including the attachment edges, has length at least four. Chaining k copies gives old-endpoint distance $L = 3k + 1$, while preserving simplicity, cubicity, connectedness, and bridgelessness.

Apply the same chain length to every edge of $G - H$, choosing $L \ge |H|$. Decompose a binary cycle avoiding M_c into edge-disjoint circuits. A circuit contained in one replacement chain uses no support edge. Every other circuit either traverses a chain from one old endpoint to the other and hence uses at least L non-support edges, or is a circuit component of H. The latter is impossible because every component of H meets M_c. Thus each remaining circuit D satisfies

$$|D - H| \ge L \ge |H| \ge |D \cap H|.$$

Summing over the circuit decomposition proves (27). □

Remark 6.2 (a one-round obstruction, not a minimum counterexample)*.* The bundled literal instance has 230 vertices, 345 edges, and support equal to two 8-circuits. Two checkers establish all four families (27): one exhausts 2^{22} base binary cycles and the other independently solves the four shortest-T-join problems in the inflated graph. The affine-class sizes are $(6, 5, 3, 2)$, the four join sizes are $(10, 11, 13, 14)$, and the four minimum containing-cycle sizes are 16. All 97 nonempty charge-restoring one-round fixed-colour path multiswitches fail to clean or delete even after every componentwise affine repair. Nevertheless, switching the $\{1, 2\}$-paths with boundary endpoints 1–10 and 4–15, then recomputing and switching the $\{1, 3\}$-path $5-6$, makes colour 2 absent from the first support circuit and permits its deletion. The graph is Tait-colourable, so its actual minimum projection is empty. The example therefore refutes neither Conjecture 4.1 nor FiveCDC. It shows that the force of global minimality must be used dynamically after each support-neutral recolouring. The proof and replay are in `scratch/general-kempe-descent-static-exchange-obstruction-20260729/`.

Proposition 6.3 (a directly clean unbounded class)**.** *Let $H =$* $\operatorname{supp}(h)$ *be one circuit. If every component W of $(V, E - H)$ contains exactly two vertices of H, then every extension $f = (h, s)$ is cleanable by independent componentwise* $\mathrm{GL}(2, 2)$ *maps followed by reintegration of the low values on H.*

Proof. At a vertex v_i of H, let $d_i \neq 0$ be the low value on the unique edge of $E - H$, and let s_{i-1}, s_i be the low values on the two incident edges of H. Flow conservation gives

$$d_i = s_{i-1} + s_i. \tag{29}$$

Summing the low-flow equations over a complement component W shows that the xor of its boundary values d_i is zero. There are exactly two, so they are equal. Fix $t \in \mathbb{F}_2^2 - \{0\}$. Independently on each W, choose a linear automorphism sending its common boundary value to t. These maps preserve the low-flow equations inside W.

The circuit H has two vertices for every complement component and hence even length. Thus (29), now with every $d_i = t$, integrates around H, and its edge values alternate between a and $a+t$. At each of the two vertices of H in W, the two support incidences therefore have one value a and one value $a+t$. Across the four incidences both values occur twice. A support edge internal to W is counted twice; deleting its two equal copies does not change parity. Consequently every low value occurs evenly on $\delta(W)$. This holds for every W, so the resulting extension is clean. □

Proposition 6.4 (conditional orbit-reflecting inflation)**.** *Let a support-boundary state realized in a simple connected bridgeless cubic graph have no clean or circuit-deletion state in one connected component of its reconfiguration graph, where the moves include all charge-preserving two-colour Kempe path multiswitches and all feasible componentwise* $\mathrm{GL}(2,2)$ *maps. There is a simple connected bridgeless cubic inflation for which:*

1. *every reachable boundary-changing move contracts to a move in the same base component, while every additional internal move contracts to the identity; and*
2. *at every reachable state, all four inequalities*

$$C \cap M_c = \varnothing \quad \Longrightarrow \quad |C \cap H| \leq |C - H| \tag{30}$$

   hold for every binary cycle C and every $c \in \mathbb{F}_2^2$.

The assertion is conditional: it does not assert that such a goal-free base component exists or that the displayed support is globally minimum.

Proof. Delete an edge ab of $K_{3,3}$, regard a, b as degree-two terminals, and attach one external edge at each terminal. Call this two-pole P. By the parity lemma, the attachment colours in every proper three-edge-colouring of P are equal, say to t; adding ab in colour t gives a Tait colouring of $K_{3,3}$, and deletion is the inverse. The two Tait-colouring classes of $K_{3,3}$ have every colour pair Hamiltonian [5, Section 4.2]. Therefore a pair containing t induces the unique terminal-to-terminal path of P, while the other pair induces an internal circuit. A switch of the first kind changes the boundary exactly as a switch on the contracted edge; a switch of the second kind leaves the boundary fixed. Both retain the two-valued internal Kempe-class label. Hence P preserves and reflects boundary-changing Kempe moves.

Replace every complement edge by a chain of r copies of P. The shortest terminal-to-terminal path inside one copy has length three, so the old-endpoint distance of the chain is $L = 4r + 1$. Equal terminal colours propagate along a chain. Thus every boundary-changing Kempe component contracts to its base component, every internal switch contracts to the identity, and every base move expands. Contraction also preserves the endpoints of simultaneous path switches and the boundary action of componentwise maps. Fresh pole vertices preserve simplicity and cubicity. Bridgelessness follows because a base circuit through a replaced edge expands through a terminal path, while every internal pole edge lies on an internal bichromatic circuit or on a terminal path completed by such a base circuit.

Choose $L \geq |H|$. In any reachable state, absence of a deletion means that every circuit component of H uses all four low values. Fix c and decompose a binary cycle C avoiding M_c into edge-disjoint circuits. A circuit internal to a replacement chain meets no support edge. Every other circuit using complement edges traverses a whole chain, so it has at least L complement edges and at most $|H|$ support edges. A circuit contained in H is impossible because every support component meets M_c. Summing over the decomposition gives (30). Orbit reflection prevents a new clean or deletion boundary state, and the same argument applies after every reachable recolouring. □

Remark 6.5 (exact replay and search boundary). The bundled dynamic Kempe frontier package contains a dependency-free enumeration of all twelve proper pole colourings, its two six-state Kempe orbits, chain distances through eight copies, and all 146,599 two-terminal pairings on one support circuit through order fourteen. A separately written checker scans all 3^9 edge assignments of $K_{3,3}$ and verifies the cut-incidence parity in Proposition 6.3. A noncanonical exploratory search over 75 larger random poles found no goal-free component; this is negative experimental evidence, not an exhaustive theorem.

The four-colour simultaneity cannot be replaced by a theorem about one matching. In the Petersen graph with edge set

$$\{03, 06, 09, 14, 16, 18, 25, 26, 27, 37, 38, 47, 49, 58, 59\},$$

the matching $M = \{06, 14, 25\}$ is contained in exactly two minimum-cardinality binary cycles:

$$\{06, 09, 14, 18, 25, 26, 49, 58\}, \qquad \{06, 09, 14, 16, 25, 27, 47, 59\}.$$

Both have size eight, and in both cases the components of the complement have M-endpoint parities $1, 0, 1$. Exhausting all 2^{15} edge subsets proves minimality. The complete finite replay is in `scratch/petersen-minimum-tjoin-countermodel-20260729/`. This obstruction does not realize four synchronized affine classes.

What is not yet understood is how the common rainbow-odd cut interacts with the four shortest-join duals. A publishable resolution would need either a cut/cycle contradiction, a neutral-reconfiguration theorem, or an exact counterexample to Conjecture 4.1. Until one of those is supplied, the correct status is a frontier result.

AI-use and verification disclosure

This research route, proof draft, checker, computation, and exposition were developed by OpenAI Codex agents under Atharva Vaidya's direction. The minimum-projection exchange proof, the size-at-most-five proof, and the span proof used at sizes six and seven are displayed in full so that they can be checked without trusting an AI system. Those boundary classifications and the stronger direct-cleaning theorem through size ten are also regenerated by dependency-free exact checkers. The through-eleven checker freezes every first direct-repair failure and its smaller-support certificate. The through-twelve package freezes all 392 new circuit-deletion certificates and includes a separately implemented full census. The matching through-thirteen packages independently reproduce all five new shapes and freeze all 7,674 deletion certificates. The size-fourteen primary package freezes all seven shape counts and the 224 residual rows. A separately written full 7 + 7 enumerator regenerates all 333 word orbits and 91,481,505 charge-valid states, while the Kempe package freezes 724 matching-robust deletion rows and checks them with an independently structured literal verifier. Codex agents also derived the universal tensor lemma, the one-circuit obstruction identity, and the circuitwise-balanced corollary displayed above. A separate hostile agent audit checked the expansion grouping, the leaf cancellation, all no-leaf graphs through seven tensor vertices, and all 4,096 three-vertex tensors; this remains agent review, not independent human verification. The size-fifteen primary package freezes all eight shape counts and 6,036 residual rows. A separately written induction checker canonicalizes every residual and links it to the matching-robust inverse-extension family. The single-circuit anchor census completed all sixteen shards and found zero failures; an independent parser checks aggregation and agreement with the zero-sum-partition recurrence, but it is not an independently implemented full candidate census. The static-exchange package contains the human proof of Proposition 6.1, a primary full-cycle checker, a separate shortest-T-join checker for the four inequalities, and the complete boundary-profile Kempe graph for its example. Separate packages also check the first failure of the unrestricted dichotomy, its 40-case map certificate, and its exhaustive 18-vertex realization analysis. The standard graph-class reduction and its D_5-label checker were likewise written by Codex agents; a separate hostile agent audited the structural proof, which is still not independent human review. The order-18 result is reproducible from canonical generation and exhaustive cycle-space enumeration. The expanded census exactly replays frozen sources but inherits their population provenance. Both remain computational results and should be independently rerun. The static-exchange proposition and example were found, formalized, and hostilely audited by Codex agents. That audit corrected the distinction between the gadget's internal-terminal distance and its old-endpoint traversal distance; it is still not independent human review. The two-terminal cleaning proof and the orbit-reflecting $K_{3,3} - e$ inflation were likewise proposed, formalized, checked, and hostilely audited by Codex agents. The dynamic package contains two independently written finite checkers, but its unbounded conclusions rest on the displayed human proofs rather than the finite tests. The $K_{3,3}$ Kempe-class fact is prior work of Belcastro and Haas.

Hušek and Šámal's flow criterion and Hoffmann-Ostenhof's matching/join formulation are prior work. This draft makes no claim of a FiveCDC resolution and no priority claim for the minimum-projection viewpoint pending expert literature review. Before public submission, a human graph theorist should check the proof line by line, rerun the census in a clean environment, audit the bibliography, and decide authorship and venue policy under the applicable AI-disclosure rules.

References

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